COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 4.1

Discrete -Time System

Aim:

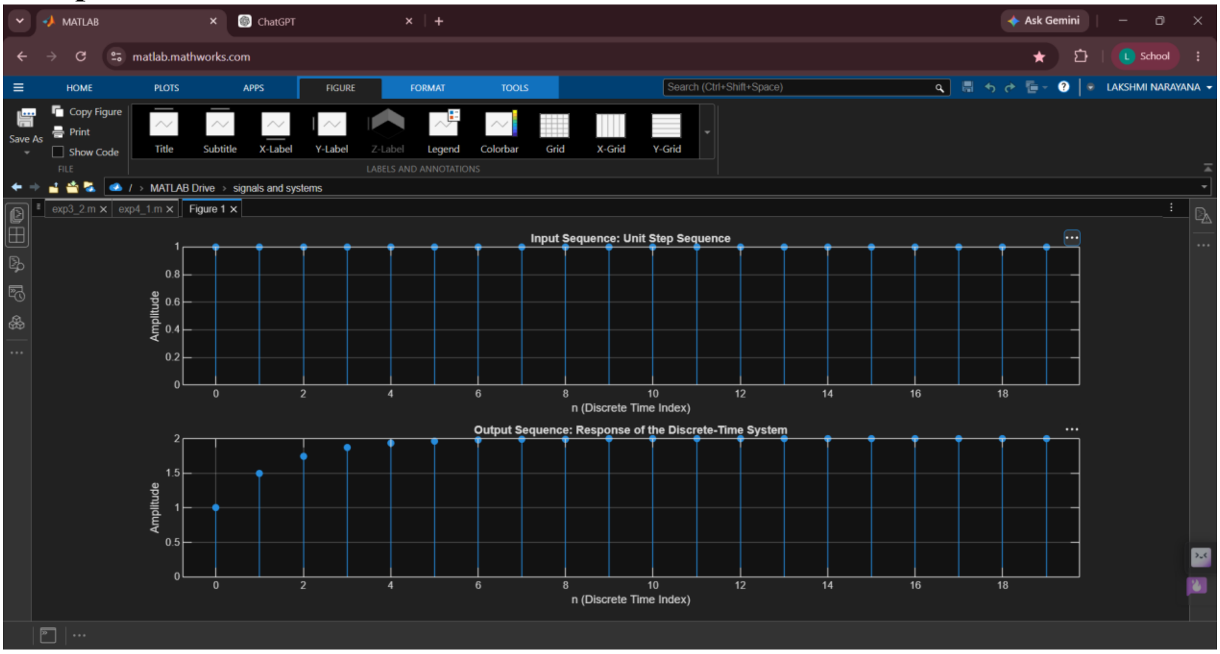
To analyze the behavior of a discrete-time linear system using a difference equation. The Aim is to understand how the input sequence is transformed by the system, represented by a difference equation, and to observe the output sequence.

Program:

```
% Test Case: Analysis of a Discrete-Time Linear System


clc;

clear;

close all;


% Define parameters

N = 20;                  % Number of samples

n = 0:N-1;                % Discrete time index vector


% Input sequence: Unit step sequence

x_n = ones(1, N);


% Output sequence initialization
```

```
y_n = zeros(1, N);

% Difference equation: y[n] = 0.5*y[n-1] + x[n]
y_n(1) = x_n(1);        % Initial condition

for k = 2:N
    y_n(k) = 0.5 * y_n(k-1) + x_n(k);
end

% Plot the input and output sequences
figure;

% Input sequence
subplot(2,1,1);
stem(n, x_n, 'filled');
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Input Sequence: Unit Step Sequence');

% Output sequence
subplot(2,1,2);
stem(n, y_n, 'filled');
grid on;
xlabel('n (Discrete Time Index)');
ylabel('Amplitude');
title('Output Sequence: Response of the Discrete-Time System');
```

Output:



Result:

- **System Behavior:** The output sequence reflects the behavior of a first-order linear system with a feedback component, where the current output depends on the previous output scaled by 0.5 and the current input.

● **Steady-State Response:** Over time, the output sequence approaches a steady state as the effect of the input dominates and the contribution from the previous output stabilizes.

This test case helps illustrate how a discrete-time linear system responds to a step input, showcasing the recursive nature of the difference equation and its effect on the output sequence.